

COLIN McLAUGHLIN

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EDUCATION

Queen's University

Expected Graduation Date: 04/2027

Bachelor of Computing, specialization in Artificial Intelligence

GPA: 3.63 | Dean's List 2023–2025

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Linear Data Analytics, Discrete Mathematics, Computer Architecture, Software Architecture, Statistics, Linear Algebra, Calculus

EXPERIENCE

AI Software Developer

June 2025 – August 2025

Ostar Education

Hybrid/Internship, Richmond Hill, ON

- Developed a full-stack AI lesson generation platform (Python, FastAPI, React, OpenAI API, PostgreSQL), building out a database, admin dashboards, role-based access, and complete lesson/user CRUD workflows that automated K–12 worksheet creation and cut teacher prep time by 75% (1 hour to 15 minutes).
- The platform supports 5+ K–12 teachers and has produced over 150 unique worksheets since deployment.
- Deployed the solution on a secure Linux cloud server, implementing DevOps best practices including CI/CD pipelines (GitHub Actions), systemd service management, Nginx reverse proxying, and automated process monitoring.

Consulting Project Lead

September 2025 – Present

QMIND x Technical Management Group Ltd.

Queen's University, Kingston, ON

- Leading a team of 5 developers to automate contractor CV formatting into a standard company template for a mining/energy consulting firm, reducing a 40-minute task to a 2-minute proofread (95% reduction).
- Building a pipeline integrating PDF/LinkedIn parsing, GPT-4, and LangChain for structured JSON output.
- Organizing stakeholder meetings and maintaining GitHub workflows to deliver a client-ready automation system.

AI Developer - Natural Language Processing

September 2024 – April 2025

QMIND

Queen's University, Kingston, ON

- Developed an electroencephalograph ML pipeline (Pandas, NumPy, scikit-learn) and trained a Convolutional Neural Network (TensorFlow) to classify emotion from brain signals; achieved 95% accuracy on select datasets.
- Produced detailed analysis notebooks and visualizations (Matplotlib), which accelerated model iteration.

Other Experience: Residence Don — Queen's University (Aug 2024 – Apr 2025); Computer Camp Counselor — Three-Flavours Kids' Club (Jun 2024 – Aug 2024); Steward — Angus Glen Golf Club (Apr 2022 – Sep 2022)

PROJECTS

 github.com/colin-mclaughlin

Visulo | Tauri, React/TypeScript, Rust, SQLite, OpenAI API/Ollama, FAISS, Tesseract (OCR), Whisper (ASR) 2025

- Developed an AI study overlay tool that captures on-screen text/regions, performs OCR to answer user queries, and includes automated note-taking (editable/exportable), progress tracking, and multimodal (voice/text) interaction.
- Built full local pipeline: Tauri desktop shell with React/TypeScript UI, Rust backend for capture and storage, e5-small embeddings via FAISS vector index, and pluggable LLM providers (OpenAI/Ollama) for RAG answers.

CodeOps Agent | Python, FastAPI, LangGraph, OpenAI/Claude, MCP Tools, FAISS, React, Docker, GitHub Actions 2025

- Developed an agentic DevOps assistant that monitors GitHub CI pipelines, retrieves semantic context from code and logs, and autonomously proposes and tests fixes in a sandboxed environment.
- Built modular Model Context Protocol (MCP) tools (GitTool, TestRunnerTool, NotifierTool) integrated with a LangGraph ReAct reasoning loop using RAG over code and CI logs via FAISS embeddings.

SKILLS

Programming Languages: Python, Java, JavaScript/TypeScript, SQL (PostgreSQL, SQLite), C, Rust, HTML/CSS

Frameworks & Libraries: React, Node.js, FastAPI, Tauri, Tailwind, LangChain, LangGraph, Pandas, NumPy, TensorFlow

Dev. Tools & AI: GitHub, Docker, GitHub Actions (CI/CD), Linux; FAISS, RAG pipelines, LLM & API integration

ACTIVITIES & ACHIEVEMENTS

- AI Research, Queen's Hyperloop Design Team; built ML routing algorithm (Python, QGIS); co-authored paper for European Hyperloop Week, won full-scale Socio-Economic Award (2024) Sept 2023 – Apr 2025
- Delivering paid work as a remote contractor on Outlier; prompt engineering, data annotation May 2025 – Present
- Queen's University Hackathon (QHacks) Competitor; built Living Journal, a full-stack AI journal app Jan 2025